

CardioVision: ECG Classification Project Report

Project Overview

This project focuses on the automated classification of electrocardiogram (ECG) signals using deep learning. We utilize the standard **MIT-BIH Arrhythmia Database**, a widely recognized benchmark for evaluating heartbeat classification algorithms. The dataset contains ECG signals categorized into 5 distinct classes, allowing the model to distinguish between normal heartbeats and various types of arrhythmias.

Architecture

The core model is the **CardioVisionHybrid**, a powerful deep neural network that combines two major architectures to effectively process sequential ECG data:

1. **1D Convolutional Neural Network (CNN)**
 - Acts as an automated feature extractor.
 - Uses sliding 1D convolutional layers (`nn.Conv1d`) to capture localized, spatial features from the raw ECG waveforms (e.g., identifying QRS complexes, P waves, and T waves).
 - Utilizes Batch Normalization, ReLU activations, and Max Pooling to progressively downsample the signal while extracting high-level representations.
2. **Long Short-Term Memory (LSTM)**
 - Acts as the temporal sequence processor.
 - The feature maps extracted by the CNN are fed sequentially into a two-layer LSTM.
 - Since ECG signals are time-series data, the LSTM is critical for understanding the rhythm and temporal dependencies across the heartbeat over time.
3. **Fully Connected Classifier**
 - The final hidden state from the LSTM is passed into a fully connected multilayer perceptron (MLP) with dropout for regularization.
 - Outputs raw logits across the 5 target classes.

Training Details

- **Hardware:** CUDA (GPU Acceleration)
- **Optimizer:** Adam Optimizer (Learning Rate: 0.001)
- **Loss Function:** Cross-Entropy Loss
- **Batch Size:** 64
- **Epochs Trained:** 35

Results & Performance

Following 35 epochs of training, the model achieved an impressive **Overall Accuracy of 98.53%** (approaching ~99%). The model's state-dictionary has been successfully persisted to `cardiovision_hybrid_weights.pth`.

Detailed Classification Report

Class	Precision	Recall	F1-Score	Support (Samples)
0 (Normal)	0.99	1.00	0.99	18,118
1 (Supraventricular)	0.90	0.81	0.85	556
2 (Ventricular)	0.96	0.95	0.96	1,448
3 (Fusion)	0.88	0.69	0.78	162
4 (Unknown)	0.99	0.99	0.99	1,608

[!TIP] While the model performs exceptionally well on the majority class (Normal beats) and Ventricular/Unknown categories, the lowest recall is observed on Class 3 (Fusion beats). Future improvements could involve techniques like synthetic data augmentation (SMOTE) or weighted loss functions to improve performance on these under-represented classes.